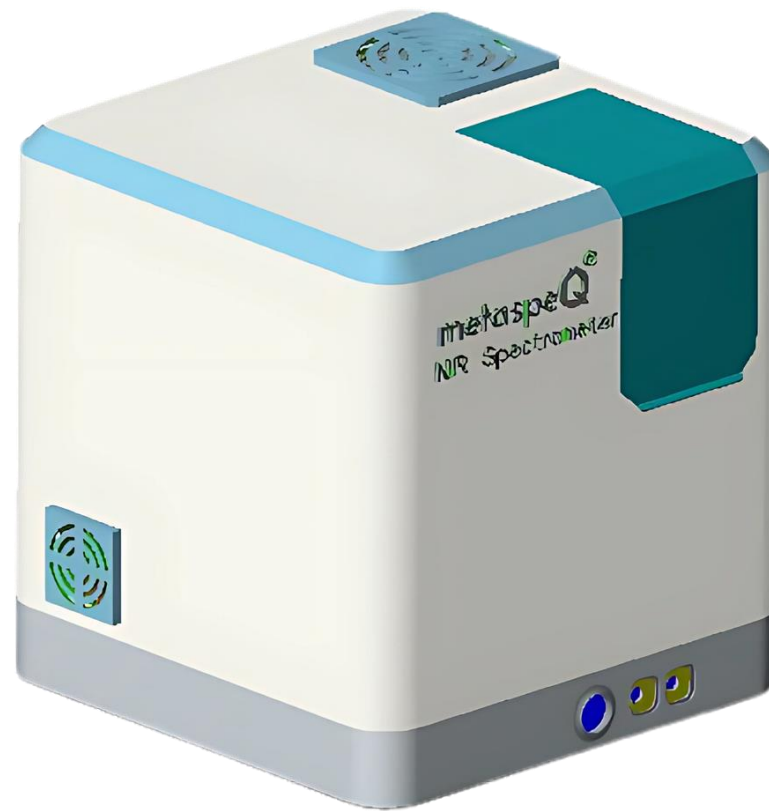




Next Generation Quality Control Solution



Revolutionizing Quality Assurance with AI-Powered NIR Spectroscopy

AYUDYOG PRIVATE LIMITED

<https://www.metaspeq.com/>

Current Scenario



mint

Markets Premium Money Mutual Fund Industry Companies Technology

Business News / News / India / WHO flags 7 India-made syrups in its probe on contamin...

WHO flags 7 India-made syrups in its probe on contaminated medicine causing 300 deaths globally

1 min read • 20 Jun 2023, 01:54 PM IST

Business Standard

Friday, April 05, 2024 | 12:41 AM IST | EN | Hindi

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65% of MSME firms found manufacturing substandard drugs since Dec: Report

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INDIA

Quality control is critical to the Indian pharmaceutical industry's future success

Mar 28, 2023, 8:28 PM IST - Anshu Jain in Varanasi, India | TOI



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Packaged Food & Drinks Delivered In India Are Of The Worst Quality Anywhere In The World

Oxford's George Institute for Global Health analysed over 400,000 food and drink products from 12 countries across the world for the study. UK topped the list with the healthiest packaged food and drinks followed by the US and Australia. Study noted that that food packages and drinks in India are most energy-dense.

Shweta Sengar — Updated: Aug 22, 2019, 23:48 IST —



Current Challenges in Quality Control:

- High costs & inefficiency in quality testing
- Lack of scalable, affordable solutions for MSMEs
- Counterfeit drugs & substandard manufacturing

Market Need:

A cost-effective, fast, and scalable testing solution

Problem

Industries like pharmaceuticals, food, agriculture and wellness lack real-time, portable and cost-effective quality assessment tools for detection of counterfeiting and adulteration as existing solutions are inefficient, expensive, complex, and require extensive expertise.

This results in poor QA adoption causing approximately 65% of MSMEs producing substandard products, leading to combined losses of upto Rs. 50,000 Cr to Indian pharma and food companies

- Packaged foods in India, loss is Rs. 20,378 crore
- Indian Pharma losses due to counterfeiting and sub-standard drugs is Rs. 35000 Cr

Source:

https://intellectual-property-helpdesk.ec.europa.eu/news-events/news/role-indian-customs-fight-against-counterfeits-2022-10-19_en

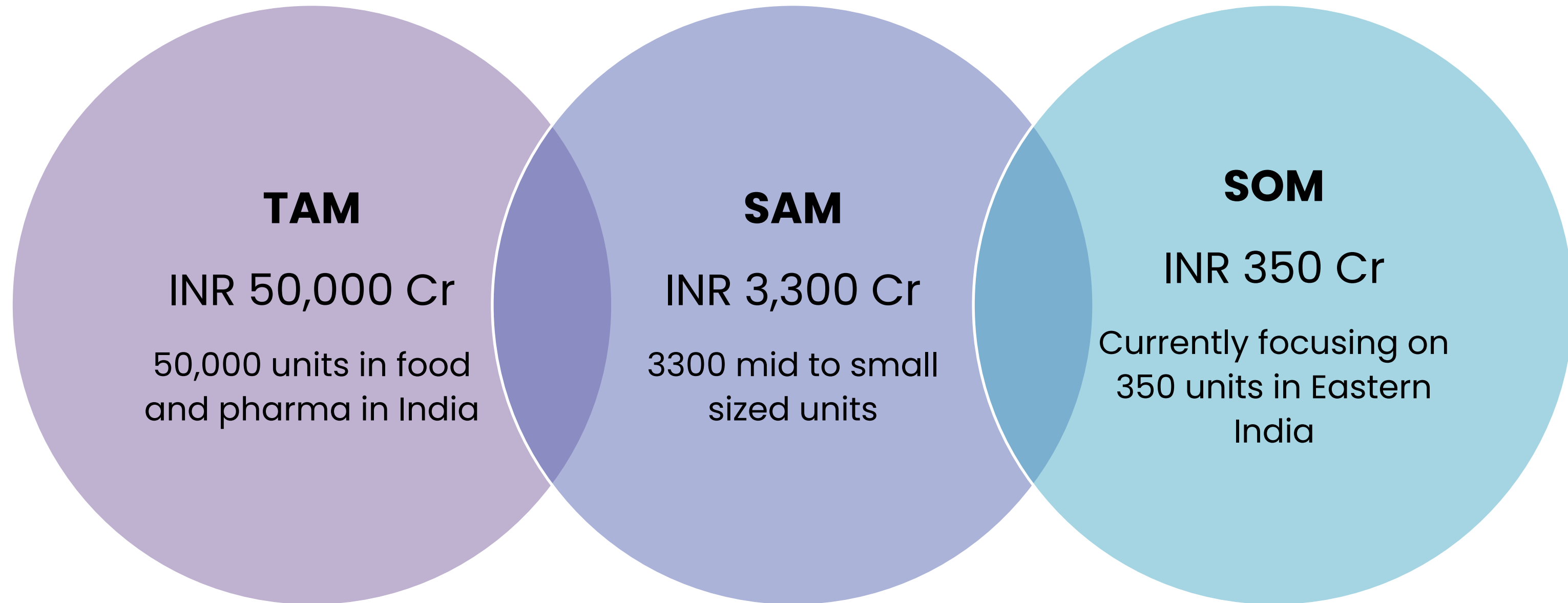
<https://journalofsocialsciences.org/vol6no1/combating-counterfeit-and-substandard-medicines-in-india--legal-framework-and-the-way-ahead/>

Our Market Research Findings

There is a strong need for Rapid, Cost-Effective Quality Assurance Solutions

Current status	Pain Point	Market Needs
Ability to stop counterfeiting	No reliable & compliant solutions	Detection of counterfeit and sub-standards product in circulation
Sub standard products	Complex procedure with Multiple steps, Extensive R&D expertise	Automated instant detection
Device costs	Very high initial as well as running costs	Cost effective solutions
Difficult to manage Regulatory Compliance	Complex & Expensive	Data driven automated process

Market Size



*Bottoms Up Market Strategy based on PMR

Solution



metaspeQ[®]

Portable desktop
NIR spectrometer



SPECTRATMFIND
AI for Spectroscopy

METASPEQ:

India's first portable NIR spectrometer for real-time quality and traceability of substances, foods, and pharmaceuticals

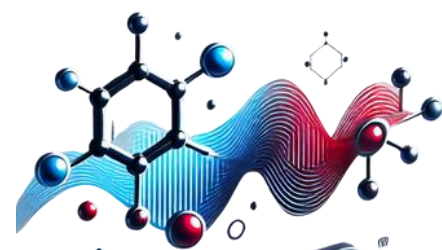
SpectraFind:

AI-powered software that automates NIR model development and validation, making spectroscopy faster, easier, and more accessible

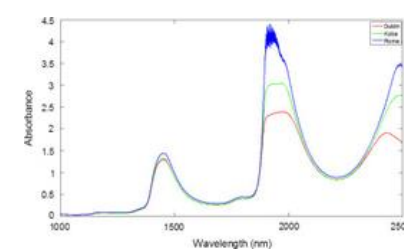
Technology



End-to-end QA solution



Sample Interaction
Molecular Absorption of NIR Radiation: Understanding Chemical Bond Interactions



Identify & Quantify
Spectral Peaks: Key to Identification and Quantification of Sample Components



AI-Powered Analysis
Machine Learning for Rapid & Clear Parameter Estimation

Value Proposition

Cost-Effective:

**10× cheaper than existing solutions,
90% savings on initial costs**

No consumables:

100% savings on running costs

Automation:

**AI-Powered SpectraFind enhances
model**

**Building and validation outcomes by
5x**

Output:

4x faster, saving time & effort

Multi-Industry Applications:

**Pharma, Ayurveda, food, feed, and
medicinal plants.**

Future Potential:

Big data analytics & AI integration

Team



Dr. Dilip Sing
Ph.D. (Engg.)
Founder, Tech Head



Dr. Subhadip Banerjee
Ph.D. (Pharma.)
Founder, Application Science Head



Prof. Rajib Bandyopadhyay
Professor, Jadavpur University
Co-Founder



Arbab Mehtab
M.Tech (Instrumentation)



Surjyoti Mondal
B.Tech (EC)



Uday Pratap Singh
M.Pharm
Founder, Sales & Marketing Head



Soumitra DasMahapatra
M.Pharm
Founder, Administration



Surojit Roy
Business Mentor
PadUp Ventures



Deba Prasad Kanshari
B.Tech (EC)



Jadavpur University



Technology
Innovation Hub
of IIT Delhi

PadUp
ventures



IIM CALCUTTA
INNOVATION PARK



West Bengal



Competitive advantage



40% Cost effective, Customized AI Solution

Feature	Ayudyog (METASPEQ)	Traditional NIR Systems
Portability	✓ Yes	✗ No
AI Automation	✓ Yes (SpectraFind)	✗ No
Customization	✓ Yes	✗ Limited
Cost	✓ Affordable (Rs. 12 lacs)	✗ Expensive
		(~ Rs. 50 lakhs to Rs. 1 Cr)
Industry Focus	✓ Multi-industry	✗ Limited
Make in India	✓ Yes	✗ No

Business Model

B2B Enterprise Model

Device Price + SAAS and AMC

- **Upfront Revenue: Hardware sale**
- **Recurring Revenue: Subscription + AMC + Customized Solution + Accessories**
 - Subscription:
 - Cloud-Based AI & Analytics
 - Monthly or annual fee for software services.
 - Accessories: Sample holder, white standard, Light source
 - AMC: Post-Warranty Support, Annual fee for maintenance & service.
- **Customized Solution: Specific model and Library**

Pricing

Device Price:

- **Desktop: INR 12 lakhs**
- **Handheld: INR 8 lakhs**
- **In-line process monitoring system: INR 50 lakhs**

SaaS (Software-as-a-Service):

- **Cost: INR 1 lacs per annum per machine**

Milestones



Traction & Validation

Initial customers:

- Spectramind Tea Solutions Private Limited
- Directorate of Cinchona and Other Medicinal Plants, Govt. of WB
- Mendine Pharmaceuticals Pvt. Ltd.
- Bengal Herbs
- Nasa remedies
- HERBS – N- DRUGS

Revenue booked: Rs. 60 lacs

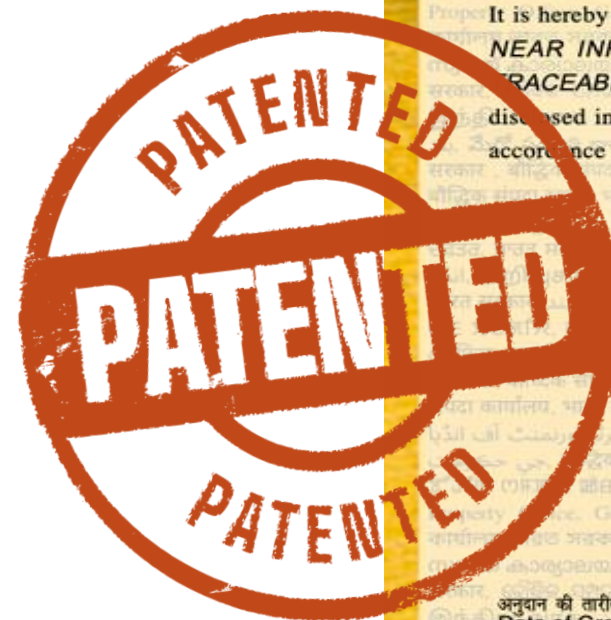
Advance received: Rs. 18 lacs

Pipeline:

**Successful demonstrations
with 40+ potential clients**

Example: East India Pharma

Intellectual Property (Patent)





**INTELLECTUAL
PROPERTY INDIA**
PATENTS & TRADE MARKS
GEOGRAPHICAL INDICATIONS



सत्यमेव जयते

क्रम सं./SL No.: 033137768



पेटेंट कार्यालय, भारत सरकार

पेटेंट प्रमाण पत्र

(पेटेंट नियमावली का नियम 74)

The Patent Office, Government Of India

Patent Certificate

(Rule 74 of The Patents Rules)

पेटेंट सं. / Patent No. : 557176

आवेदन सं. / Application No. : 202331057717

फाइल करने की तारीख / Date of Filing : 28/08/2023

पेटेंटी / Patentee : Ayudhyog Private limited

आविष्कारकों का नाम / Name of Inventor(s) : 1.Dr. Dilip Sing 2.Dr. Subhadip Banerjee 3.Prof. Rajib Bandyopadhyay 4.Mr. Kalyan Majumdar

पेटेंट सं./SL No.: 033137768

पेटेंट प्रमाण पत्र

(पेटेंट नियमावली का नियम 74)

The Patent Office, Government Of India

Patent Certificate

(Rule 74 of The Patents Rules)

प्रमाणित किया जाता है कि पेटेंटी को, उपरोक्त आवेदन में यथाप्रकटित **A PORTABLE NEAR INFRA-RED (NIR) SPECTROMETER FOR ESTIMATING THE QUALITY AND THE TRACEABILITY OF NATURAL SUBSTANCES LIKE CINCHONA AND THE METHOD THEREOF** नामक आविष्कार के लिए, पेटेंट अधिनियम, 1970 के उपबंधों के अनुसार आज तारीख अगस्त 2023 के अट्ठाईसवें दिन से बीस वर्ष की अवधि के लिए पेटेंट अनुदत्त किया गया है।

It is hereby certified that a patent has been granted to the patentee for an invention entitled **A PORTABLE NEAR INFRA-RED (NIR) SPECTROMETER FOR ESTIMATING THE QUALITY AND THE TRACEABILITY OF NATURAL SUBSTANCES LIKE CINCHONA AND THE METHOD THEREOF** as disclosed in the above mentioned application for the term of 20 years from the 28th day of August 2023 in accordance with the provisions of the Patents Act, 1970.

अनुदान की तारीख / Date of Grant : 31/12/2024



पेटेंट नियंत्रक / Controller of Patents

(Signature)

टिप्पणी - इस पेटेंट के नवीकरण के लिए फीस, यदि इसे बनाए रखा जाना है, अगस्त 2025 के अट्ठाईसवें दिन को और उसके पर्याप्त प्रत्येक वर्ष में उसी दिन देय होगी।

Note. - The fees for renewal of this patent, if it is to be maintained, will fall / has fallen due on 28th day of August 2025 and on the same day in every year thereafter.

Financial Overview & Unit Economics

Revenue in Cr

Year	1st Year (FY 2025–26)	2nd Year (FY 2026–27)	3rd Year (FY 2027–28)
Revenue	7.50	24.50	60.66

Expenses in Cr

Year	1st Year	2nd Year	3rd Year
Total Direct Cost	3.90	15.00	32.80
Total Indirect Cost	4.40	5.80	5.49
Gross Margins	48%	39%	46%

Ask

**Seeking INR 2.5 Crores
for 7.5% dilution**

- **Scale production and deployment.**
- **Expand sales and marketing efforts.**
- **Enhance AI-driven analytics and cloud capabilities.**

Fund Utilization (%)

Category	Allocation (%)
Working Capital	40%
Sales & Marketing	30%
Operational & Team Expansion	20%
R&D and AI Enhancements	10%
Total	100%



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